

Visual acuity and cognitive outcomes at 4 years of age in a double-blind, randomized trial of long-chain polyunsaturated fatty acid-supplemented in...

BACKGROUND: While there is a large body of data on the effects of long-chain polyunsaturated fatty acid supplementation of infant formula on visual and cognitive maturation during infancy, longterm visual and cognitive outcome data from randomized trials are scarce. **AIM:** To evaluate docosahexaenoic acid (DHA) and arachidonic acid (ARA)-supplementation of infant formula on visual and cognitive outcomes at 4 years of age. **METHODS:** Fifty-two of 79 healthy term infants who were enrolled in a single-center, double-blind, randomized clinical trial of DHA and ARA supplementation of infant formula were available for follow-up at 4 years of age. In addition, 32 breast-fed infants served as a "gold standard". Outcome measures were visual acuity and the Wechsler Preschool and Primary Scale of Intelligence--Revised. **RESULTS:** At 4 years, the control formula group had poorer visual acuity than the breast-fed group; the DHA- and DHA+ARA-supplemented groups did not differ significantly from the breast-fed group. The control formula and DHA-supplemented groups had Verbal IQ scores poorer than the breast-fed group. **CONCLUSION:** DHA and ARA-supplementation of infant formula supports visual acuity and IQ maturation similar to that of breast-fed infants.